

Capstone Project Final Report Guidelines

The Capstone Project may contain 40-60 pages (excluding coding) with double spacing. The project documentation must be with respect to the project only. Capstone Project/Mini Project Report should strictly follow the points given below:

Report Formatting

The information given in this section explains the formatting of the report. Use any word processing tools/latex to set them up.

Layout

- Use A4 size with 35mm left margin 30mm right, top, and bottom margins
- Page numbers should be included (See Tables of Content).

Fonts

- Font type: Times New Roman
- Font Size: 12pt

Alignments

- Allow 1.15mm line-spacing
- Paragraphs should be separated by one blank line
- Permitted font color - only black
- Left and right margins should be justified

Legends

- Should be emboldened and centered

Page limit (Quantity can't substitute the Quality)

- There are no any limitations/strict rules about overall page count.
- Get advice from your respective supervisor regarding the page limit

Print

- The whole report should be printed on one side
- Type: Grayscale

Structure

- Follow the instructions given in Anatomy of the Report

Anatomy of the Report

Title Page

Generally, the title page contains a precise title, names, and affiliations of investigators along with institutional details. Don't do fancy though you are free to follow your own set of ideas.

Declaration

Type the particular declaration given in *Appendix A*.

Certificate of Approval

Appendix B.

Acknowledgment

Even though it is optional, the gratitude towards the people who gave the real contribution and support can be praised here (typically your supervisor, head of the department, friends and colleagues, etc.).

Abstract

The abstract is the shortened form or the summary of the complete report. Give more emphasis on results, recommendations, and conclusion, and make it more concise. Apart from these major points, the purpose of the report can also be included. This section;

- can contain 250 - 300 words

- should be written as a single paragraph using the present tense

Table of Contents

The arrangement of the complete report can be illustrated in the table of contents. Use Roman numbering except for primary chapters, for which the numerals can be used.

List of Figures

The list of figures with numbers and titles can be shown together with page numbers where the figures are located. Keep in mind that all figures and tables listed here must be referred to inside the main text.

E.g. "The use case diagram shown in Figure 1"

List of Tables

This list of all tables can be shown together with their page numbers. The arrangement of tables should be organized by chapter.

Chapter 1: Introduction

Through this chapter, a detailed description of the project can be given to the audience. The major points elaborated in other chapters of the report have to be merged together with a brief description. Most importantly, it should encourage the reader to read the complete report.

This section may include the following major points as subsections;

- ✓ Major goals and objectives
- ✓ Motivation
- ✓ The scope of the completed project
- ✓ The approach and assumptions while carrying out the project work
- ✓ Concise summary of major outcomes

The length of this chapter depends on the nature of the project, where 4000-8000 words is adequate in most cases

Chapter2: Background

The background section should provide the reasons why the project was initiated, which helps the audience gain a complete understanding of the rest of the report. The major attention has to be given in explaining the reason why the project is intended to address the problem pointed out in the report. With that, a clear indication of other related works where the same problem has been tried previously.

Chapter 3: Specification and Design

A specification should describe what the software system is required to do once it is implemented. In more simple words, the specification can be referred to as “expectation” and described as “what the proposed software system does.” The design always gives the description about top-level details of how the software requirements are met.

A strong recommendation is to make extensive use of;

- ✓ Use case diagrams
- ✓ ER diagrams
- ✓ UML diagrams
- ✓ Sequence diagrams
- ✓ State charts
- ✓ Algorithms

Chapter 4: Methodology

In the Methodology, provide more description at a finer level of detail by outlining the method of implementation, including the Machine Learning/Deep Learning pipeline, down to the coding level of your project until the deployment.

Describe any problems that caused difficulties during implementation with the solution you found in solving them. Try to describe critical code segments in the system, significant interfaces, and other components.

Methodology may include;

- ✓ Data Collection and Preprocessing
- ✓ Feature Engineering and Selection
- ✓ Model Selection, Training and Testing

- ✓ Model Development and Integration
- ✓ Software and hardware requirements
- ✓ Illustration of a non-standard or innovative way of implementing an algorithm and data structure
- ✓ Difficulties involving existing software
- ✓ Lack of appropriate supporting software
- ✓ Over-ambitious project aims

Chapter 5: Results and Evaluation

The primary purpose of the results and evaluation is to indicate the inferences drawn from your observations. Pay more attention to describing how you demonstrated that the system works as intended.

In evaluating the results, you may include:

- ✓ The comparison of experimental results with expected values
- ✓ A description of the interrelationship of the experimental results
- ✓ Analyze and state the achieved accuracy
- ✓ Analyze and state the implications or limitations

Chapter 6: Future Work

Some discussions and critiques of your project, in order to provide openings to future research, should be included in this section. This can be provided as an opportunity for future researchers by expressing unrevealed ideas.

In this section, you may include;

- ✓ Gaps of the project
- ✓ Proposal for Enhancement or Re-Design

Chapter 7: Conclusions

The conclusion chapter should state the achievements of the project briefly. These should be derived directly from the results and evaluation. Simply, it is referred to as the summary of the findings and may include;

- ✓ The importance of the result
- ✓ Validity of the result
- ✓ Gaps and limitations of the findings

References

List down all important references which you have obtained the information or ideas in preparing the project report. You may include the consulted list of all books, research articles, and technical resources. Each and every reference should be cited in the main text. Follow IEEE standards for referencing.

Glossary

A list of special technical words or acronyms may be necessary. This is particularly true if the subject deals with a new area with specialized vocabulary that the average reader in the discipline might not be familiar with, such as Biotechnology. This list should come after the appendices.

Appendix A - Sample Declaration Page Declaration (18pt + B)

I declare that this report does not incorporate, without acknowledgment, any material previously submitted for a Degree or a Diploma in any University, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person or ourself except where due reference is made in the text. Also, I hereby grant Sabaragamuwa University of Sri Lanka the non-exclusive right to reproduce and distribute my report, in whole or in part, in print, electronic, or other media. I retain the right to use this content in whole or in part in future works (such as articles or books).

Index Number

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Name of Student

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Date

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Signature of Student

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Appendix B – Certificate of Approval

We hereby declare that this report is the student's own work and effort, and that all other sources of information used have been acknowledged. This report has been submitted with our approval.

Name of Internal Supervisor:

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Signature of Internal Supervisor

Date:

Name of Internal Co-Supervisor :

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Signature of Internal Co-Supervisor

Date:

Name of Head of the Department:

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Signature of Head of the Department

Date: